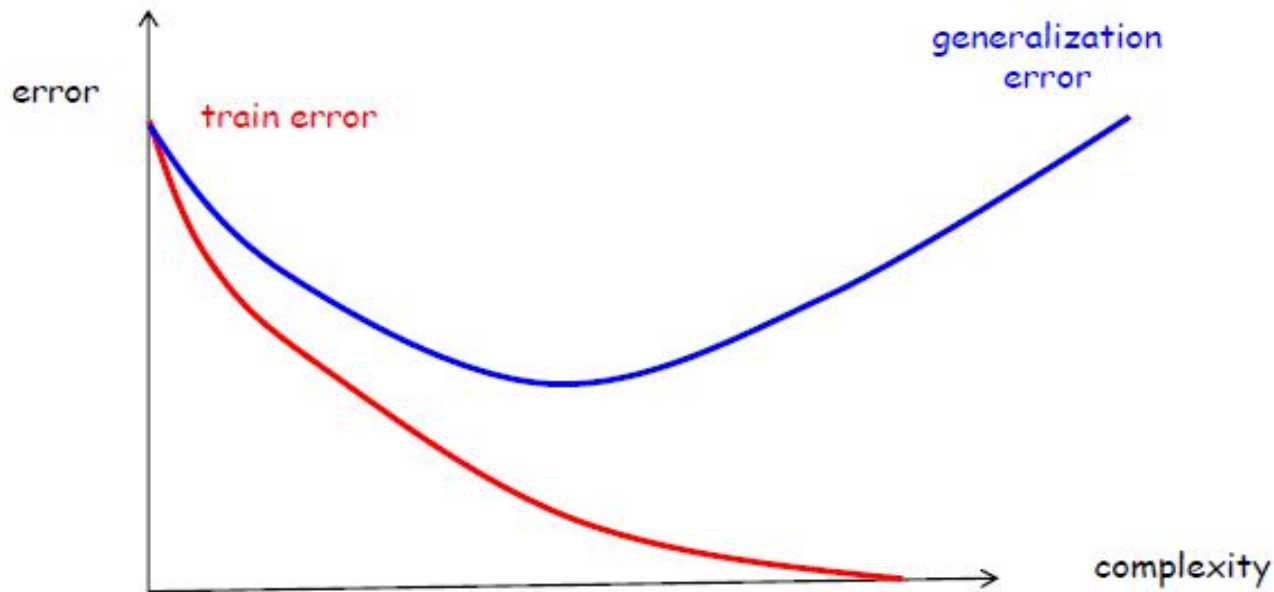


Model Selection and Validation



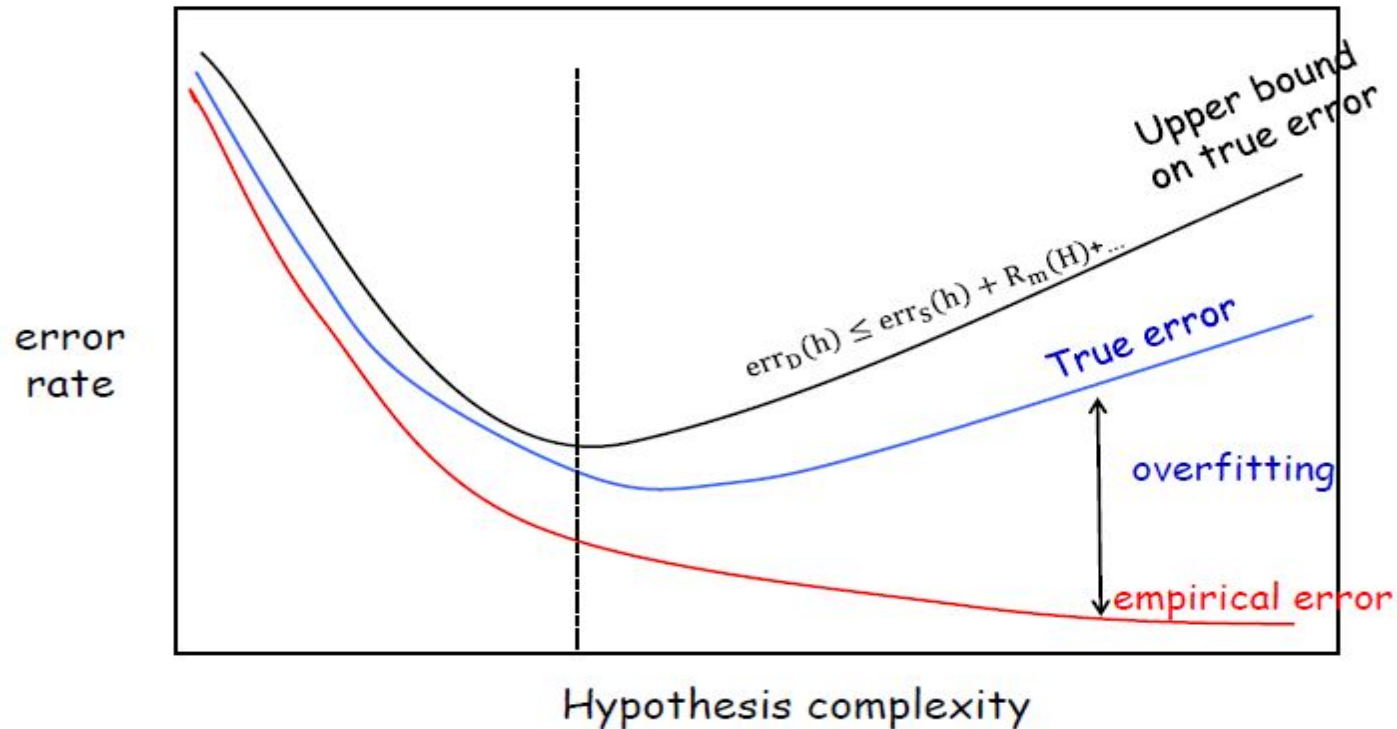
Can we use bounds for model selection?

- ▶ True error vs. Training error vs. Overfitting
- ▶ Tradeoff between decreasing training error and keeping Hypothesis complexity simple (model complexity)



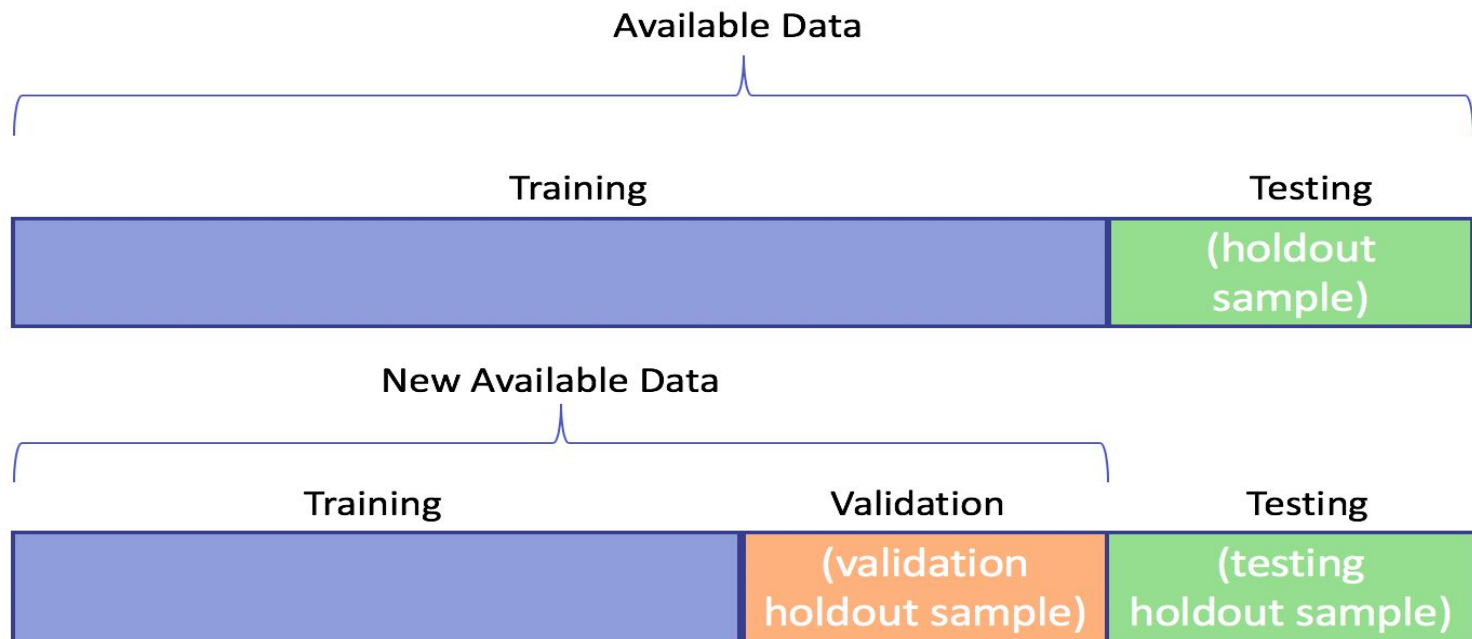
Structural Risk Minimization (SRM)

$$H_1 \subseteq H_2 \subseteq H_3 \subseteq \dots \subseteq H_i \subseteq \dots$$



Train, Test and validation split

- ▶ More training data gives better generalization.
- ▶ More test data gives better estimate for the classification error probability.
- ▶ Never evaluate performance on training data. The conclusion would be optimistically biased.



Objective of Validation

- ▶ **Algorithm selection:**
 - ▶ Selecting the *class of models* which is best-suited for the data at hand (tree-based models vs neural networks vs linear models)
- ▶ **Hyperparameter tuning:**
 - ▶ Tuning the hyperparameters of a model to *increase* this predictive power
- ▶ **Measure of generalizability:**
 - ▶ Computing an *unbiased* estimate of the predictive power of the model

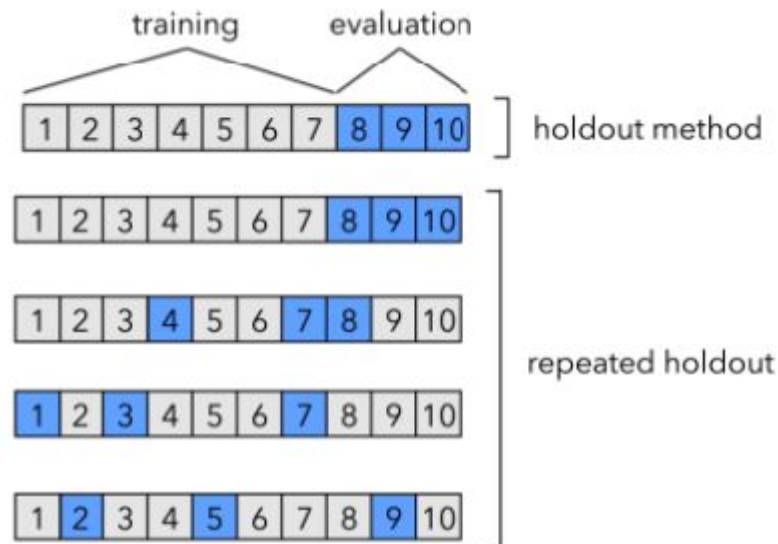
Note: First two objective comes under **model selection** and third one refer as **model evaluation**.

Partitioning the data

- ▶ Hold out
- ▶ Cross validation
- ▶ Bootstrap

Hold out method

- ▶ Given data is randomly partitioned into two independent multisets.
 - ▶ Training multiset (e.g., 2/3 of data) for the statistical model construction, i.e. learning the classifier.
 - ▶ Test multiset (e.g., 1/3 of data) is hold out for the accuracy estimation of the classifier.
- ▶ Random sampling is a variation of the hold out method:
 - ▶ Repeat the hold out k times, the accuracy is estimated as the average of the accuracies obtained.



K-fold cross validation

- ▶ The training multiset is randomly divided into K disjoint multisets of equal size where each part has roughly the same class distribution.
- ▶ The classifier is trained K times, each time with a different multiset held out as a test multiset.
- ▶ The estimated error is the mean of these K errors.



LOOCV: Leave-one-out cross validation

- ▶ A special case of K -fold cross validation with $K = n$, where n is the total number of samples in the training multiset.
- ▶ n experiments are performed using $n - 1$ samples for training and the remaining sample for testing.
- ▶ It is rather computationally expensive.
- ▶ Leave-one-out cross-validation does not guarantee the same class distribution in training and test data!

Bootstrap

- ▶ The bootstrap uses sampling with replacement to form the training multiset.
 - ▶ Given: the training multiset T consisting of n entries.
- ▶ Bootstrap generates m new datasets T_i each of size $n' < n$ by sampling T uniformly with replacement. The consequence is that some entries can be repeated in T_i .
- ▶ In a special case (called 632 boosting) when $n' = n$, for large n , T_i is expected to have $1 - 1/e = 63.2\%$ of unique samples. The rest are duplicates.
- ▶ The m statistical models (e.g., classifiers, regressors) are learned using the above m bootstrap samples.
- ▶ The statistical models are combined, e.g. by averaging the output (for regression) or by voting (for classification).